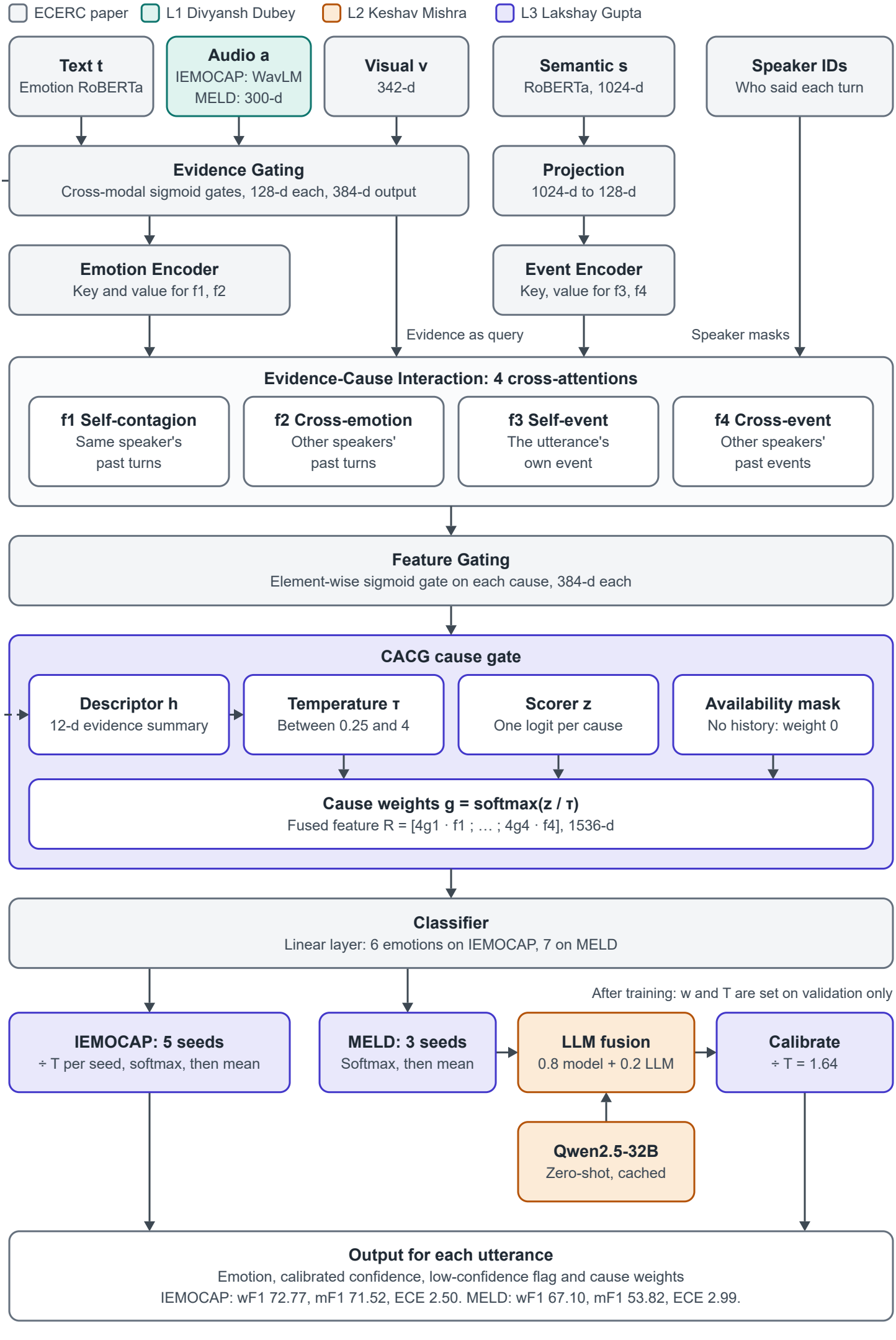

Approach B: all three limitations in one rewritten codebase
ECERC paper
L1 Divyansh Dubey
L2 Keshav Mishra
L3 Lakshay Gupta
Text t
Emotion RoBERTa
Audio a
IEMOCAP: WavLM
MELD: 300-d
Visual v
342-d
Semantic s
RoBERTa, 1024-d
Speaker IDs
Who said each turn
Evidence Gating
Cross-modal sigmoid gates, 128-d each, 384-d output
Projection
1024-d to 128-d
Emotion Encoder
Key and value for f1, f2
Event Encoder
Key, value for f3, f4
Evidence as query
Speaker masks
Evidence-Cause Interaction: 4 cross-attentions
f1 Self-contagion
Same speaker's past turns
f2 Cross-emotion
Other speakers' past turns
f3 Self-event
The utterance's own event
f4 Cross-event
Other speakers' past events
Feature Gating
Element-wise sigmoid gate on each cause, 384-d each
CACG cause gate
Descriptor h
12-d evidence summary
Temperature τ
Between 0.25 and 4
Scorer z
One logit per cause
Availability mask
No history: weight 0
Cause weights g = softmax(z / τ)
Fused feature R = [4g1 · f1 ; … ; 4g4 · f4], 1536-d
Classifier
Linear layer: 6 emotions on IEMOCAP, 7 on MELD
After training: w and T are set on validation only
IEMOCAP: 5 seeds
÷ T per seed, softmax, then mean
MELD: 3 seeds
Softmax, then mean
LLM fusion
0.8 model + 0.2 LLM
Calibrate
÷ T = 1.64
Qwen2.5-32B
Zero-shot, cached
Output for each utterance
Emotion, calibrated confidence, low-confidence flag and cause weights
IEMOCAP: wF1 72.77, mF1 71.52, ECE 2.50. MELD: wF1 67.10, mF1 53.82, ECE 2.99.